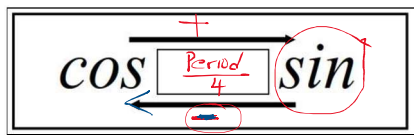
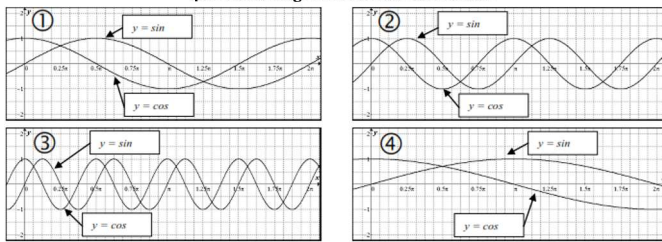


4.2 Converting between Sine and Cosine



Q1. Convert sin to cos or cos to sin in each of the following.

#	a	P	d	c	Equation with sin	Same curve using cos
a)	1	2π	$\frac{\pi}{4}$ left	0	$y = \sin(x + \frac{\pi}{4})$	$y = \cos(x + \frac{\pi}{4} - \frac{\pi}{2})$ $y = \cos(x - \frac{\pi}{4})$
b)	2	4π	$\frac{3\pi}{2}$ left	2 up	$y = 2 \sin[\frac{1}{2}(x + \frac{3\pi}{2})] + 2$	$y = 2 \cos[\frac{1}{2}(x + \frac{3\pi}{2} - \frac{\pi}{2})] + 2$
c)	7	π	$\frac{\pi}{4}$ left	1 down		
d)	$\frac{1}{2}$	$\frac{\pi}{4}$	$\frac{\pi}{8}$ right	$\frac{3}{5}$ up		
e)	3	8π	$\frac{5\pi}{6}$ right	$\frac{1}{8}$ down		

f) $y = \sin(\frac{1}{8}x)$

$y = \cos[\frac{1}{8}(x - 4\pi)]$

g) $y = 2\cos(\frac{3}{4}x)$

$y = 2 \sin[\frac{3}{4}(x + \frac{2\pi}{3})]$

$P = \frac{2\pi}{k}$
 $= \frac{2\pi}{3/4}$
 $= \frac{8\pi}{3}$
 $\frac{P}{4} = \frac{8\pi}{3} \div 4$
 $= \frac{2\pi}{3}$
 $\frac{P}{4} = \frac{2\pi}{3}$

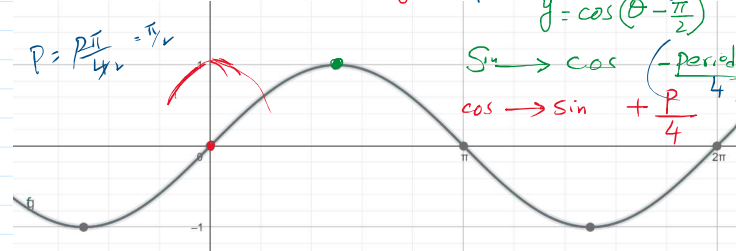
pg. 17 $\frac{P}{4} = 4\pi$

$y = A \cos[k(x-d)] + c$

$y = \sin \theta$

$y = \cos(\theta - \frac{\pi}{2})$

$\sin \rightarrow \cos(-\frac{\text{period}}{4})$
 $\cos \rightarrow \sin + \frac{P}{4}$



$k = \frac{2\pi}{P}$
 $= \frac{2\pi}{2\pi}$
 $= 1$
 $k = \frac{2\pi}{P}$
 $= \frac{1}{2}$

$\frac{P}{4} = \frac{2\pi}{4} = \frac{\pi}{2}$

$y = 2 \cos[\frac{1}{2}(x + \frac{\pi}{2})] + 2$
 $\frac{P}{4} = \frac{4\pi}{4}$

h) $y = \sin 4 \left(x - \frac{\pi}{8} \right)$ _____

i) $y = 2 \sin \frac{1}{3} \left(x - \frac{7\pi}{3} \right)$ _____

j) $y = 3 \cos 3 \left(x + \frac{\pi}{4} \right)$ _____

Q2. Write an equation to represent a i) Sine function, ii) Cosine function with the following properties. Maximum: 23 Minimum: 11 Period 5 Phase shift: 9

$$y = A \sin [k(x-d)] + C$$

$$A = \frac{\text{max} - \text{min}}{2} = \frac{23 - 11}{2} = 6$$

$$C = \frac{\text{max} + \text{min}}{2} = \frac{23 + 11}{2} = 17$$

$$d = 9$$

$$k = \frac{2\pi}{P}$$

$k = \frac{2\pi}{5}$

$f(x) = 6 \sin \left[\frac{2\pi}{5} (x - 9) \right] + 17$

$y = 6 \cos \left[\frac{2\pi}{5} \left(x - 9 - \frac{5}{4} \right) \right] + 17$

$y = 6 \cos \left[\frac{2\pi}{5} \left(x - \frac{41}{4} \right) \right] + 17$

$\frac{P}{4} = \frac{5}{4}$

Q3. Find an equation of a function of the form $y = \cos(\theta) + q$ whose graph passes through the point $A \left(\frac{\pi}{3}, -\frac{1}{2} \right)$.

$q = -1$

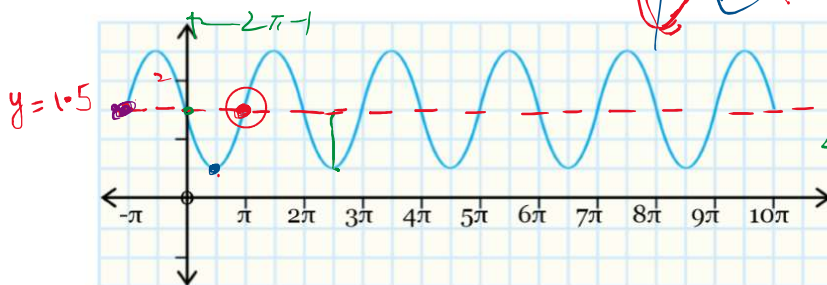
$y = \cos \theta - 1$

$$y = C$$

Q.4: State the amplitude and period, phase shift, and vertical shift of each function. Then, write the equation of the function as :

a) a sine function $y = a \sin[k(x-d)] + C$

identify max, min
a, C, k, P



+cos: max
-cos: min

+sin: pt. on Axis of curve y
-sin: " " " " " y
choos pt's closer to y-axis

$$\begin{aligned} a &= 1 \\ C &= 1.5 \\ d &= \pi \\ k &= \frac{2\pi}{P} \end{aligned}$$

$$k = \frac{2\pi}{2\pi}$$

$$k = 1$$

$$y = \sin(x - \pi) + 1.5$$

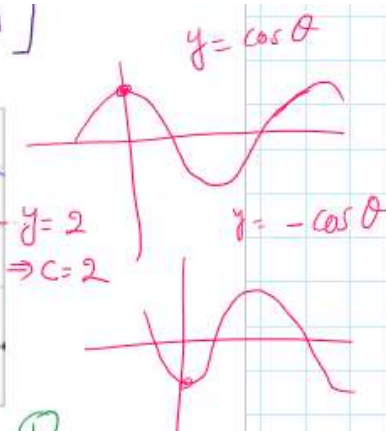
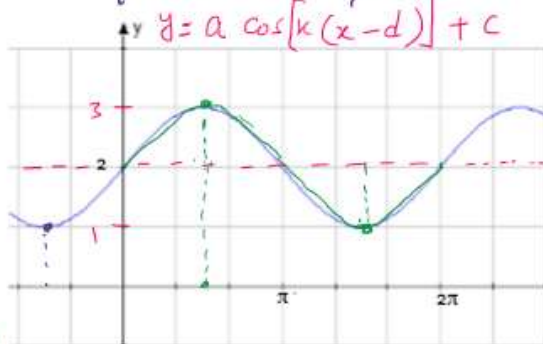
$$y = \sin(x + \pi) + 1.5$$

$$y = -\sin x + 1.5$$

$$y = -\cos(x - \frac{\pi}{2}) + 1.5$$

b) a cosine function:

b) a cosine function: [choose max./min pt's]



$$\begin{aligned} a &= 1 \\ C &= 2 \\ k &= \frac{2\pi}{\text{Period}} \\ &= \frac{2\pi}{2\pi} \end{aligned}$$

$$k = 1$$

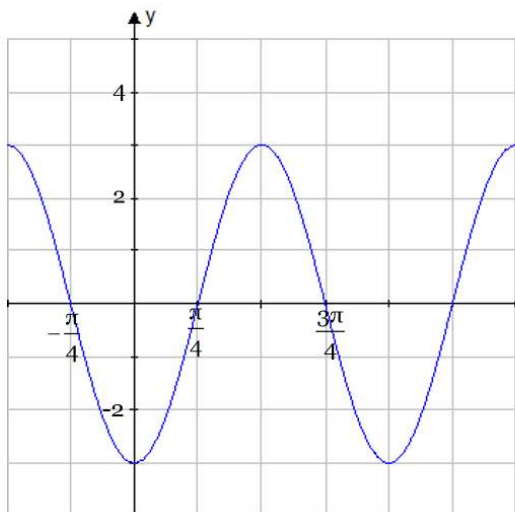
$$y = \cos(x - \frac{\pi}{2}) + 2$$

$$y = -\cos(x + \frac{\pi}{2}) + 2$$

Note: ① For + cosine eq. Choose max
② For - cosine eq. Choose min

One possible answer

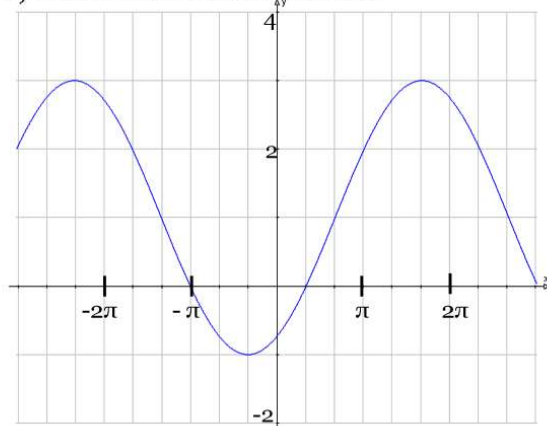
c) A cosine function.



$$y = 3 \cos \left[2 \left(x - \frac{\pi}{2} \right) \right]$$

One possible answer

d) A convenient sinusoidal function.



$$y = -2 \sin \left[\frac{1}{2} \left(x + 4 \frac{\pi}{3} \right) \right] + 1$$